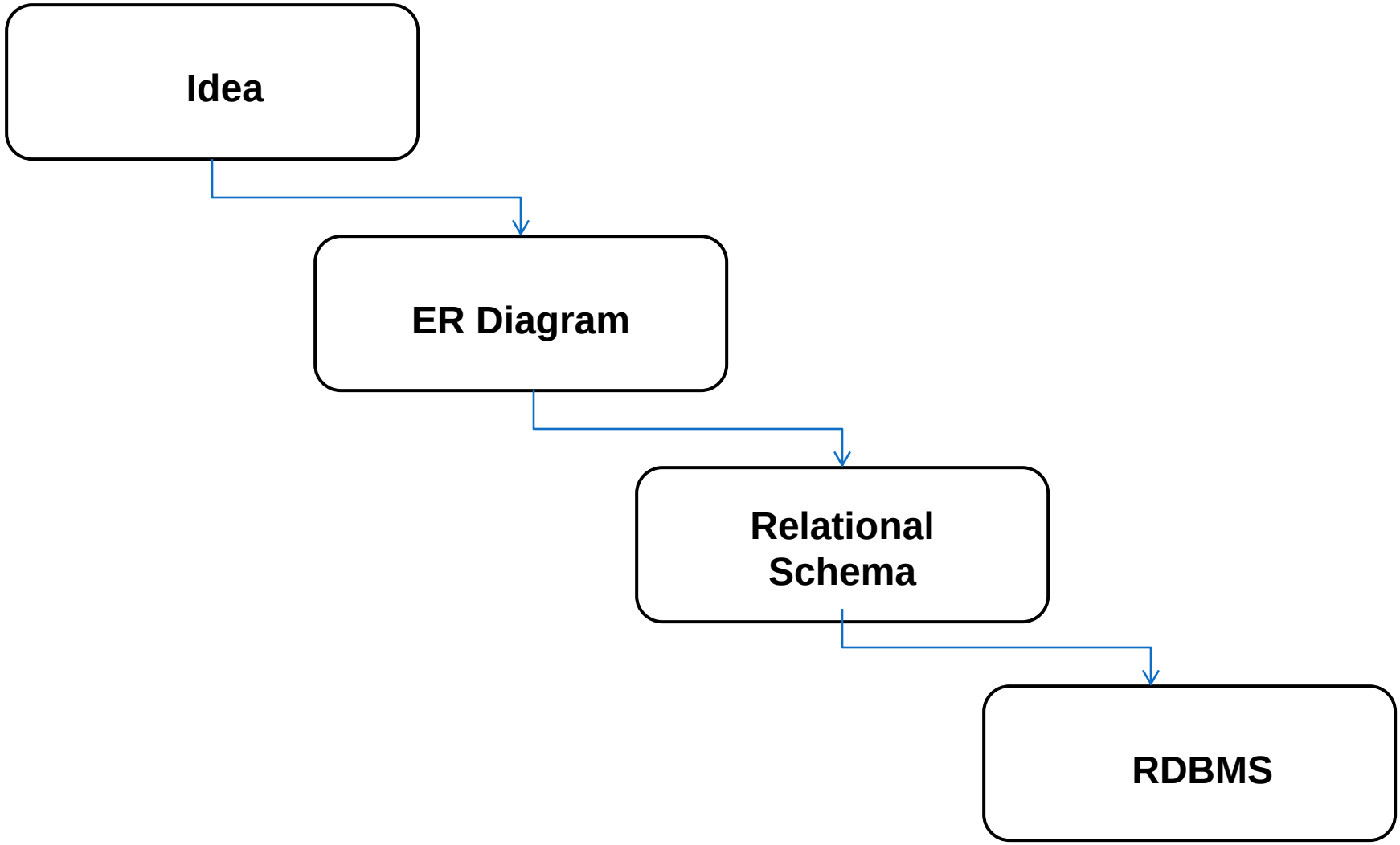
# CS105.3
# Database Management Systems

# ER Diagrams

**By: Manoja Weerasekara & Chalani Oruthotaarachchi**

# Database designing using ER diagrams

# Steps…



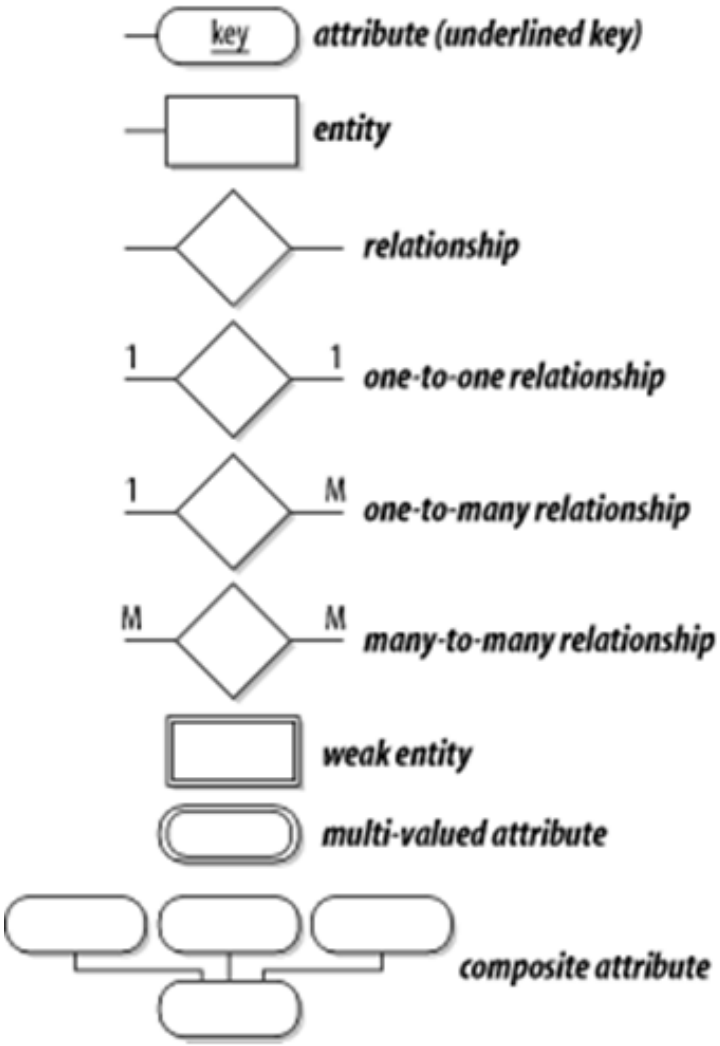
In a university, a Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course.

# ERD



- The major activity of this phase is identifying entities, attributes, and their relationships to construct model using the Entity Relationship Diagram.
  - Entity → table
  - Attribute → column
  - Relationship → line

# Chen Notation



# Classes of attributes

- Simple attribute
- Composite attribute
- Derived attributes
- Single-valued attribute
- Multi-valued attribute

# Simple/Composite attribute

- A simple attribute cannot be subdivided.
  - Examples: Age, Gender, and Marital status
- A composite attribute can be further subdivided to yield additional attributes.
  - Examples:
    - ADDRESS --→ Street, City, State, Zip
    - PHONE NUMBER --→ Area code, Exchange number

# Derived attribute

- is not physically stored within the database
- instead, it is derived by using an algorithm.
  - Example 1: Late Charge of 2%
    - MS Access: InvoiceAmt * 0.02
  - Example 2: AGE can be derived from the date of birth and the current date.
    - MS Access: int(Date() – Emp_Dob)/365)

# Single-valued attribute

- can have only a single (atomic) value.
  - Examples:
    - A person can have only one social security number.
    - A manufactured part can have only one serial number.
  - **A single-valued attribute is not necessarily a simple attribute.**
    - Part No: CA-08-02-189935
    - Location: CA, Factory#:08, shift#: 02, part#: 189935

# Multi-valued attributes

- can have many values.
  - Examples:
    - A person may have several college degrees.
    - A household may have several phones with different numbers
    - A car color

# Exercise - "Movie Database"

- Identify the types of attributes of the Movie Star Entity
  - Entity:
    - Movie Star
  - Attributes:
    - SS# : "123-45-6789"
    - Cell Phone : "(661)123-4567, (661)234-5678"
    - Name : "Harrison Ford"
    - Address : "123, Main Str., LA, CA"
    - Gender : "Female"
    - Birth date : "02-05-1990"
    - Age : "24"

# Key Attributes

* An attribute of an entity type for which each entity must have a unique value is called a **key attribute** of the entity type.

E.g. SSN of EMPLOYEE.

* A key attribute may be composite.

E.g. VehicleTagNumber is a key of the CAR entity type with components (Number, State).

* An entity type may have more than one key. For example, the CAR entity type may have two keys:
  * VehicleIdentificationNumber (popularly called VIN) and
  * VehicleTagNumber (Number, State), also known as license_plate number.

# Concepts of Keys

- Candidate key
- Primary key
- Super Key
- Alternate key/Secondary Keys
- Composite key
- Foreign key

# Candidate key

- A candidate key is a **minimal set** of attributes necessary to identify a tuple.

E.g. employeeID is a candidate key

- Each table may have one or more candidate keys. One of these candidate keys is selected as the table primary key.

| S ID | SName | Marks |
|---|---|---|
| S1 | A | 40 |
| S2 | A | 40 |
| S3 | B | 50 |
| S2 | B | 50 |

# Primary key

- One of the candidate keys that has no NULL values.
  - Null = Unknown

- An attribute (or combination of attributes) that uniquely identifies each row in a relation.

–Employee(Emp_No, Emp_Name, Department)

# Primary key-Example

**Employee**

| E-No | E-Name | D-No |
|---|---|---|
| 179 | Silva | 7 |
| 857 | Perera | 4 |
| 342 | Dias | 7 |

Primary Key

**Salary**

| E-No | Eff-Date | Amt |
|---|---|---|
| 179 | 1/1/98 | 8000 |
| 857 | 3/7/94 | 9000 |
| 179 | 1/6/97 | 7000 |
| 342 | 28/1/97 | 7500 |

←—Primary Key—→

# Composite Key

- A primary key that consists of more than one attribute

–Salary(Emp_No, Eff_Date, Amount)

# Alternate key

- An alternate key is any candidate key that is not the primary key. Alternate keys are sometimes referred to as secondary keys.
  - Null values are allowed
  - More that one alternate keys are allowed

| S ID | Bank ACC | Name | Votee ID |
|---|---|---|---|
| 1 | Acc1 | A | V1 |
| 2 | Acc2 | B | V2 |
| 3 | Acc3 | A | V3 |
| 4 | Acc4 | B | V4 |
| 5 | Acc5 | A | null |

# Super key

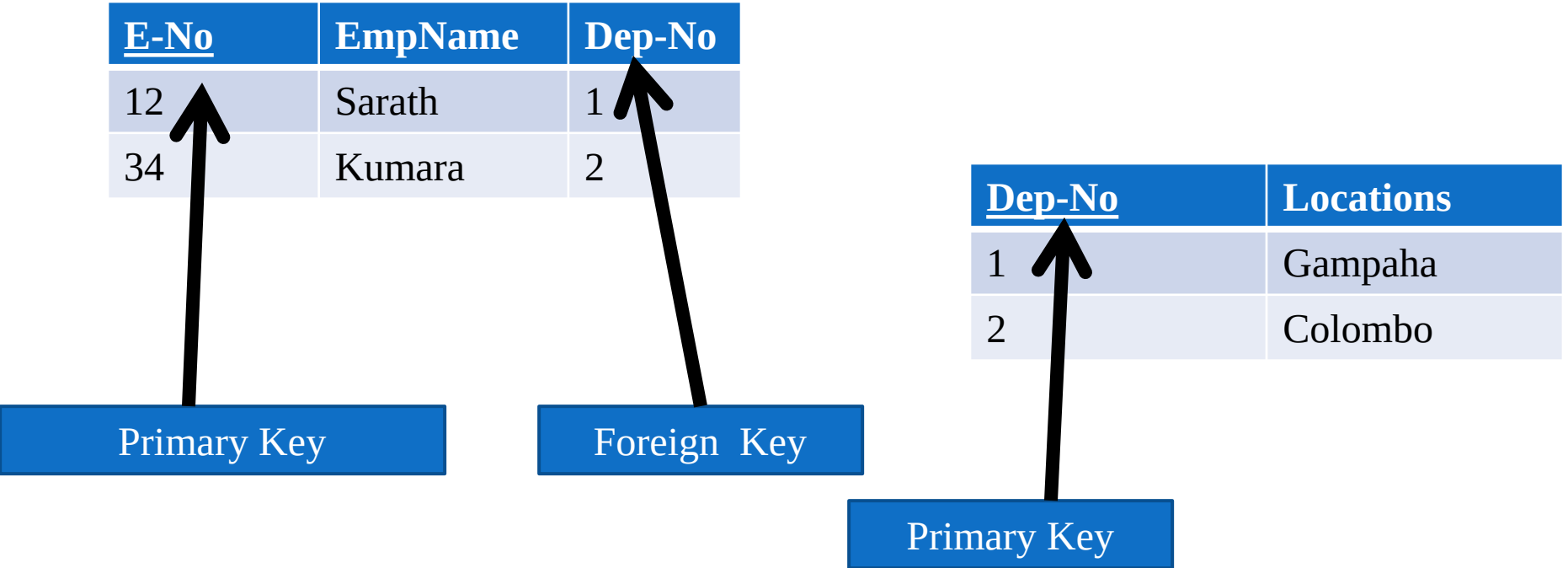
- Super key = candidate key + 0 or more attributes
  - Every candidate key is a super key
  - Every super key cannot be a candidate key

**Minimal super key becomes the candidate key**

| E ID | Salary | E Name |
|---|---|---|
| 1 | 400 | A |
| 2 | 500 | B |
| 3 | 400 | A |
| 4 | 500 | C |

# Foreign Key

An attribute in a relation of a database that serves as the primary key of another relation in the same database



# Exercise

**Database name: Ch03_Theater**

**Table name: DIRECTOR**

| DIR_NUM | DIR_LNAME | DIR_DOB |
|---|---|---|
| 100 | Broadway | 12-Jan-65 |
| 101 | Hollywoody | 18-Nov-53 |
| 102 | Goofy | 21-Jun-62 |

**Table name: PLAY**

| PLAY_CODE | PLAY_NAME | DIR_NUM |
|---|---|---|
| 1001 | Cat On a Cold, Bare Roof | 102 |
| 1002 | Hold the Mayo, Pass the Bread | 101 |
| 1003 | I Never Promised You Coffee | 102 |
| 1004 | Silly Putty Goes To Washington | 100 |
| 1005 | See No Sound, Hear No Sight | 101 |
| 1006 | Starstruck in Biloxi | 102 |
| 1007 | Stranger In Parrot Ice | 101 |

- Using the above tables identify
  - Candidate keys, primary keys for each relation.
  - alternate keys, and super keys for one relation.

# Exercise

- Below tables are extracted from employee database of an organization. Identify the primary keys and the foreign keys of each table.

**Table name: EMPLOYEE**

| EMP_CODE | EMP_LNAME | JOB_CODE |
|---|---|---|
| 14 | Rudell | 2 |
| 15 | McDade | 1 |
| 16 | Ruellardo | 1 |
| 17 | Smith | 3 |
| 20 | Smith | 2 |

**Table name: BENEFIT**

| EMP_CODE | PLAN_CODE |
|---|---|
| 15 | 2 |
| 15 | 3 |
| 16 | 1 |
| 17 | 1 |
| 17 | 3 |
| 17 | 4 |
| 20 | 3 |

**Table name: JOB**

| JOB_CODE | JOB_DESCRIPTION |
|---|---|
| 1 | Clerical |
| 2 | Technical |
| 3 | Managerial |

**Table name: PLAN**

| PLAN_CODE | PLAN_DESCRIPTION |
|---|---|
| 1 | Term life |
| 2 | Stock purchase |
| 3 | Long-term disability |
| 4 | Dental |

# Relational Integrity Constraints

- Entity integrity constraints
  - All primary key entries are unique, and no part of a primary key may be null.
- Domain integrity constraints
  - All the values of an attribute must bound to a specific range.
- Referential integrity constraints
  - Referencing attribute should be a subset of referred attribute.

# Exercise

- For each table, identify the primary key and the foreign key(s).
- Do the tables exhibit entity integrity? Answer yes or no, and then explain your answer.
- Do the tables exhibit referential integrity? Answer yes or no, and then explain your answer

**Table name: EMPLOYEE** **Database name:**

| EMP_CODE | EMP_TITLE | EMP_LNAME | EMP_FNAME | EMP_INITIAL | EMP_DOB | STORE_CODE |
|---|---|---|---|---|---|---|
| 1 | Mr. | Williamson | John | W | 21-May-64 | 3 |
| 2 | Ms. | Ratula | Nancy | | 09-Feb-69 | 2 |
| 3 | Ms. | Greenboro | Lottie | R | 02-Oct-61 | 4 |
| 4 | Mrs. | Rumpersfro | Jennie | S | 01-Jun-71 | 5 |
| 5 | Mr. | Smith | Robert | L | 23-Nov-59 | 3 |
| 6 | Mr. | Renselaer | Cary | A | 25-Dec-65 | 1 |
| 7 | Mr. | Ogallo | Roberto | S | 31-Jul-62 | 3 |
| 8 | Ms. | Johnsson | Elizabeth | I | 10-Sep-68 | 1 |
| 9 | Mr. | Eindsmar | Jack | W | 19-Apr-55 | 2 |
| 10 | Mrs. | Jones | Rose | R | 06-Mar-66 | 4 |
| 11 | Mr. | Broderick | Tom | | 21-Oct-72 | 3 |
| 12 | Mr. | Washington | Alan | Y | 08-Sep-74 | 2 |
| 13 | Mr. | Smith | Peter | N | 25-Aug-64 | 3 |
| 14 | Ms. | Smith | Sherry | H | 25-May-66 | 4 |
| 15 | Mr. | Olenko | Howard | U | 24-May-64 | 5 |
| 16 | Mr. | Archialo | Barry | V | 03-Sep-60 | 5 |
| 17 | Ms. | Grimaldo | Jeanine | K | 12-Nov-70 | 4 |
| 18 | Mr. | Rosenberg | Andrew | D | 24-Jan-71 | 4 |
| 19 | Mr. | Rosten | Peter | F | 03-Oct-68 | 4 |
| 20 | Mr. | Mckee | Robert | S | 06-Mar-70 | 1 |
| 21 | Ms. | Baumann | Jennifer | A | 11-Dec-74 | 3 |

**Table name: STORE**

| STORE_CODE | STORE_NAME | STORE_YTD_SALES | REGION_CODE | EMP_CODE |
|---|---|---|---|---|
| 1 | Access Junction | 1003455.76 | 2 | 8 |
| 2 | Database Corner | 1421987.39 | 2 | 12 |
| 3 | Tuple Charge | 986783.22 | 1 | 7 |
| 4 | Attribute Alley | 944568.56 | 2 | 3 |
| 5 | Primary Key Point | 2930098.45 | 1 | 15 |

**Table name: REGION**

| REGION_CODE | REGION_DESCRIPT |
|---|---|
| 1 | East |
| 2 | West |

# How to find relationships?

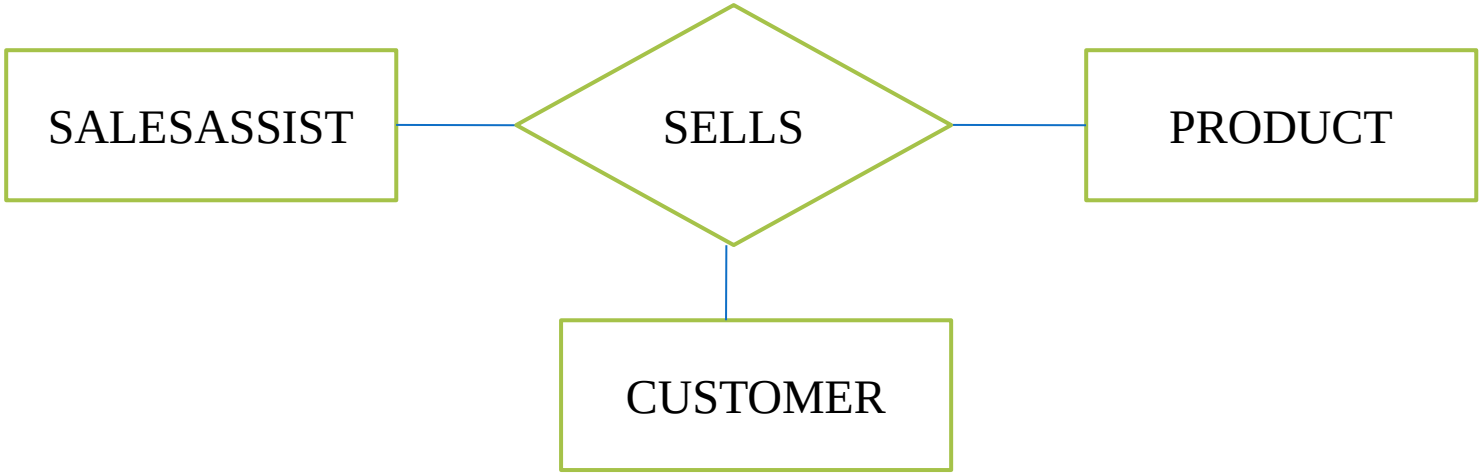


- Relationship:
  - Relationships are associations between entities.
  - Typically, a relationship is indicated by a verb connecting two or more entities.
  - Employees are assigned to projects
  - Relationships should be classified in terms of cardinality.
    - One-to-one, one-to-many, etc.

# DEGREE OF A RELATIONSHIP

It is the number of entity types that participate in a relationship

- If there are two entity types involved it is a ***binary*** relationship type
- If there are three entity types involved it is a ***ternary*** relationship type
- It is possible to have a n-array relationship

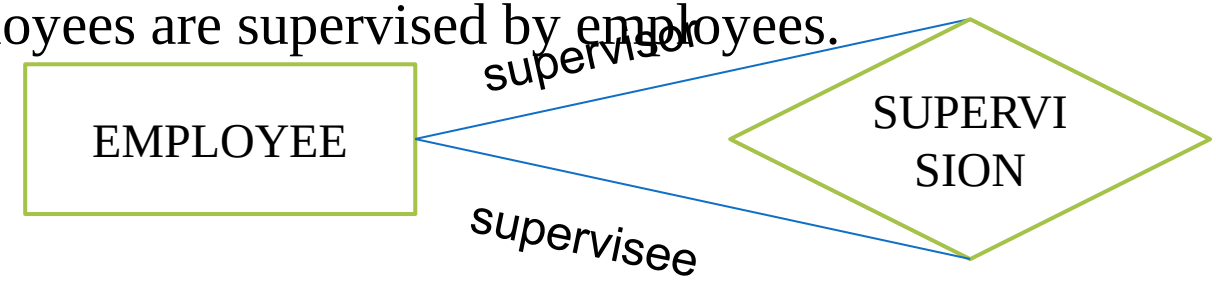


- ***Unary relationships*** are also known as a ***recursive relationship.***

- It is a relationship where the same entity participates more than once in different roles.

- In ER diagram, need to display role names to distinguish participations.

E.g. employees are supervised by employees.



# CARDINALITY CONSTRAINTS

CARDINALITY: The number of instances of one entity that can or must be associated with each instance of another entity.

- If we have two entity types A and B, the cardinality constraint specifies the number of instances of entity B that can (or must) be associated with entity A.

- Four possible categories are

  ***One to one (1:1) relationship***

  ***One to many (1:m) relationship***

  ***Many to one (m:1) relationship***

  ***Many to many (m:n) relationship***

- one-to-one



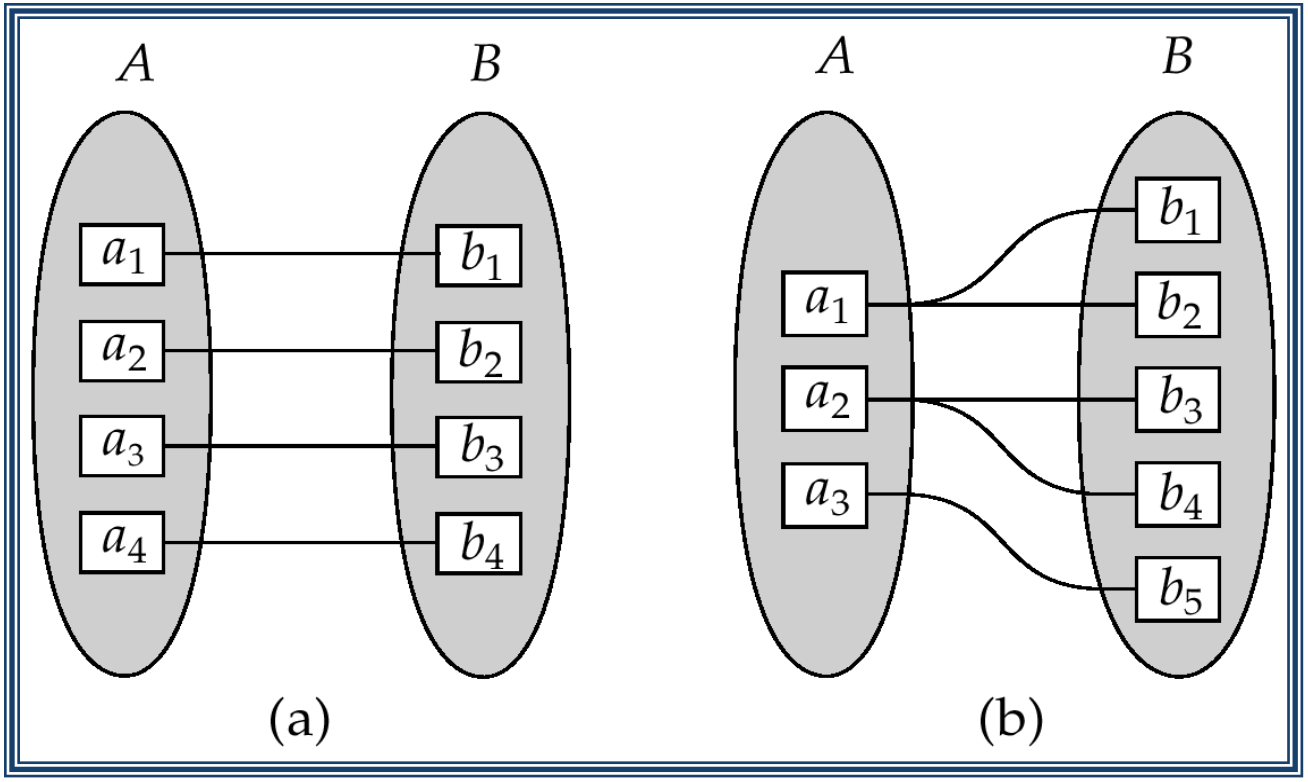

- Many to one



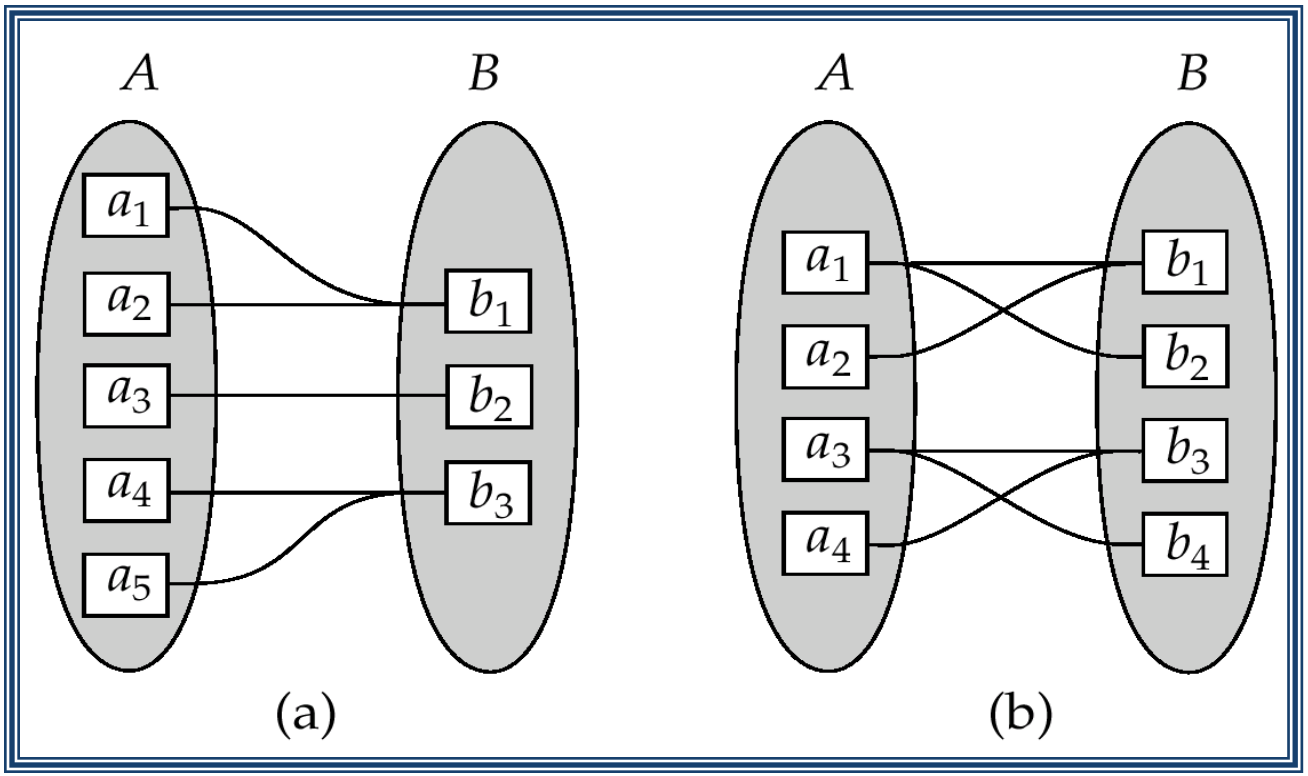

- many-to-many



# One-One and One-Many



# Many-one and many-many



# Exercise

- Using the STUDENT and PROFESSOR tables given below create an ERD.

**Database name: Ch03_CollegeQue**

**Table name: STUDENT**

| STU_CODE | PROF_CODE |
|---|---|
| 100278 | |
| 128569 | 2 |
| 512272 | 4 |
| 531235 | 2 |
| 531268 | |
| 553427 | 1 |

**Table name: PROFESSOR**

| PROF_CODE | DEPT_CODE |
|---|---|
| 1 | 2 |
| 2 | 6 |
| 3 | 6 |
| 4 | 4 |

# Weak Entity Types

* An entity that does not have a key attribute
* A weak entity must participate in an identifying relationship type with an owner or identifying entity type
* Entities are identified by the combination of:
  * A partial key of the weak entity type
  * The particular entity they are related to in the identifying entity type

- Ex: if company policy says that every employee must work for the department then participation of employee in work-for is total.




- Total participation is also called ***existence dependencies***.

- But we can't say that every employee must MANAGE a department .

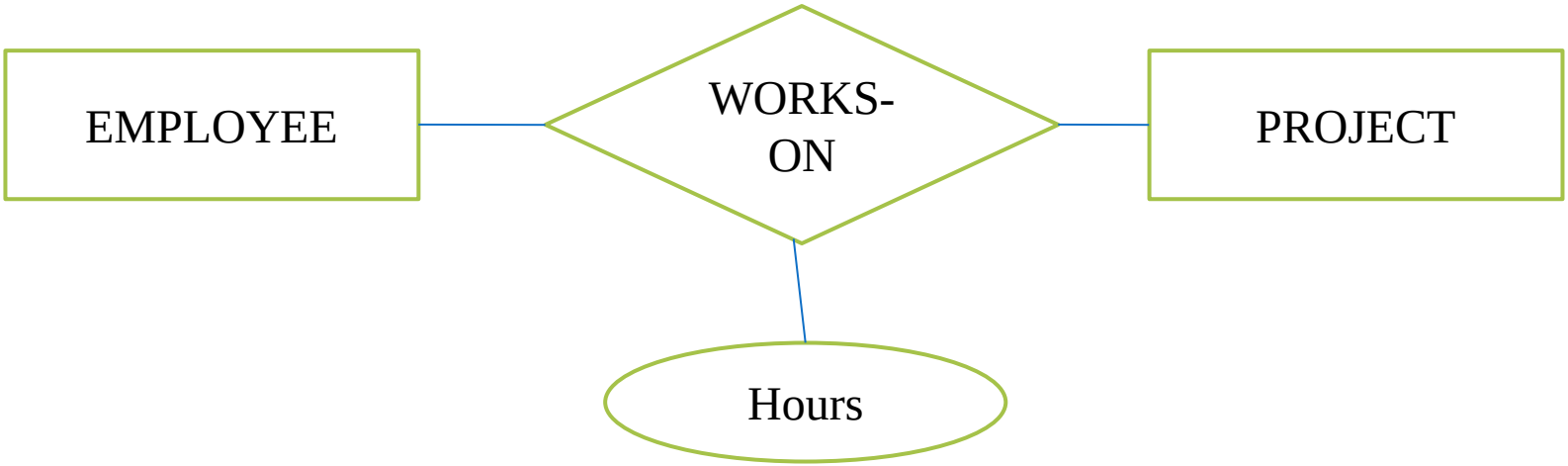
  Hence relationship is ***partial***.

- Total participation is indicated by double line and partial participation by single line.

# ATTRIBUTE OF RELATIONSHIP TYPE

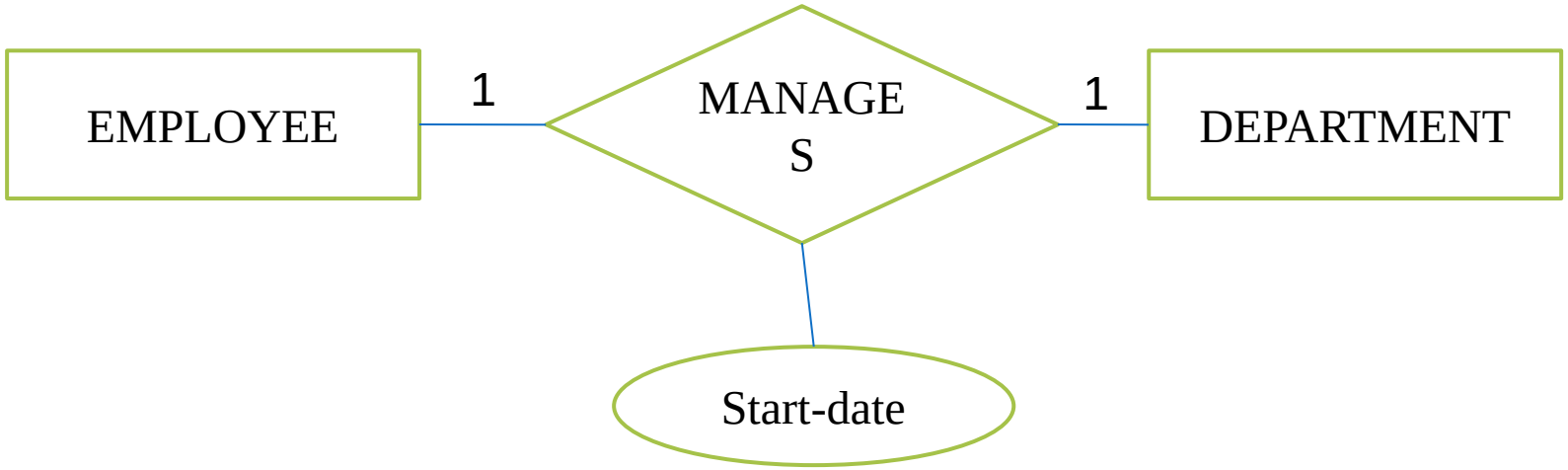
- Relationship can also have attributes

Ex: Hours for WORKS-ON relationship between EMPLOYEE and PROJECT



- Attributes of 1:1 or 1:N relationship ***can be migrated to one of the participating entity types***.

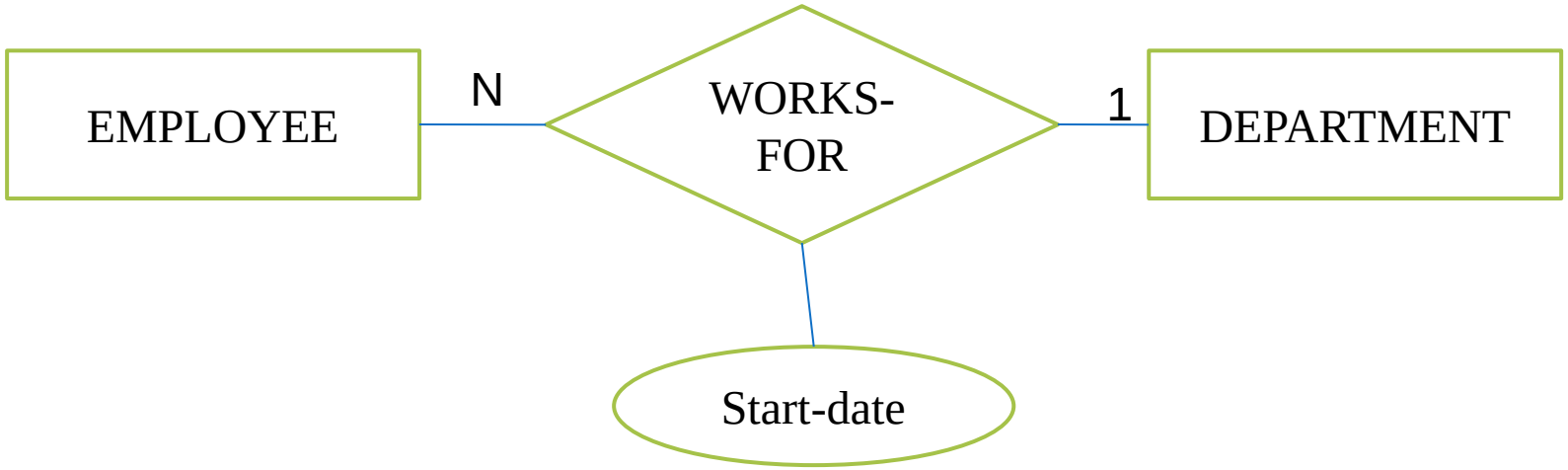
- Ex: Start-date attributes of MANAGES can be attribute of either DEPARTMENT or EMPLOYEE though conceptually it belongs to manages.



- Because each EMPLOYEE MANAGES is a 1:1 relationship.
- So every DEPARTMENT /EMPLOYEE entity participate in atmost one relationship instance.
- So value of the Start-date can be determined separately either by participating DEPARTMENT entity or participating EMPLOYEE entity.

- For 1:N relationship a relationship attribute can be migrated **only to entity type on N-side of relationship**

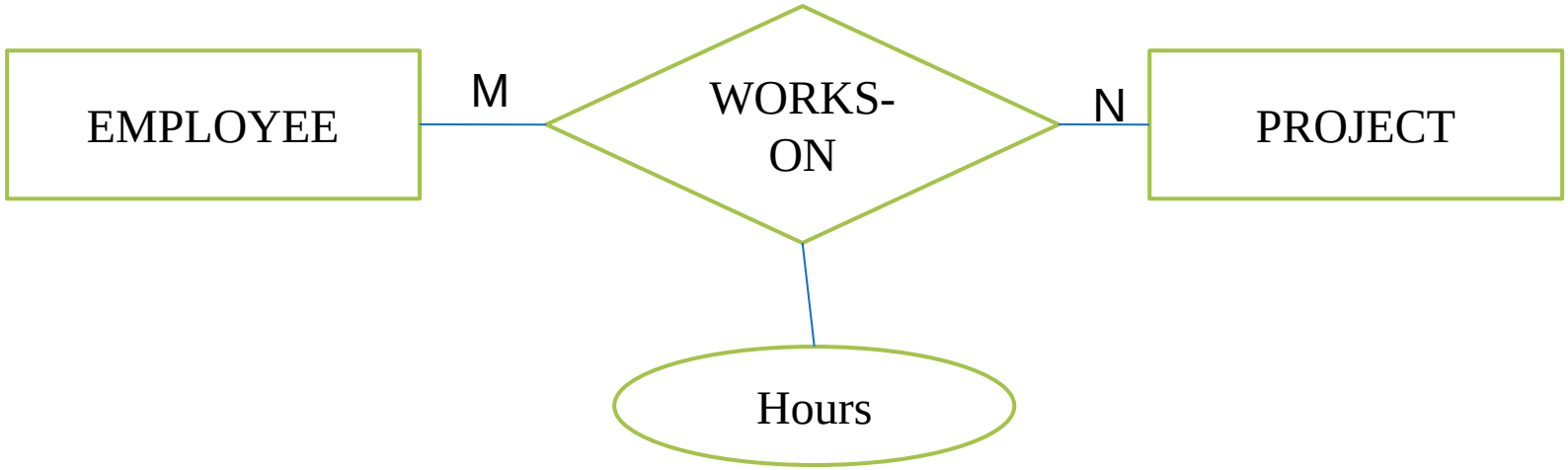
- Start-date attribute here can added only to employee.



- For M:N relationship types some attribute are determined by the combination of the participating entities, not by a single enity.

- Such attribute ***must be*** specified as the relationship attributes

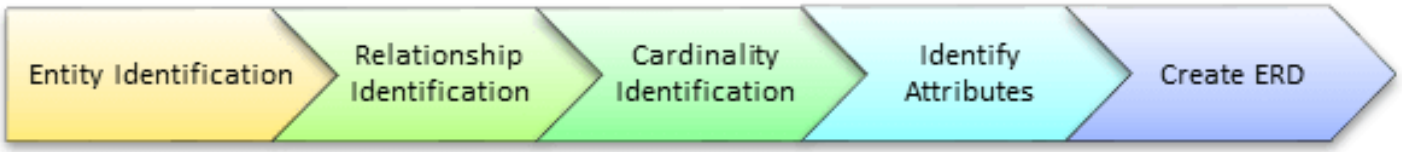
- Ex: No.of hours an employee works on is department is determent is determined by the EMPLOYEE-PROJECT combination.



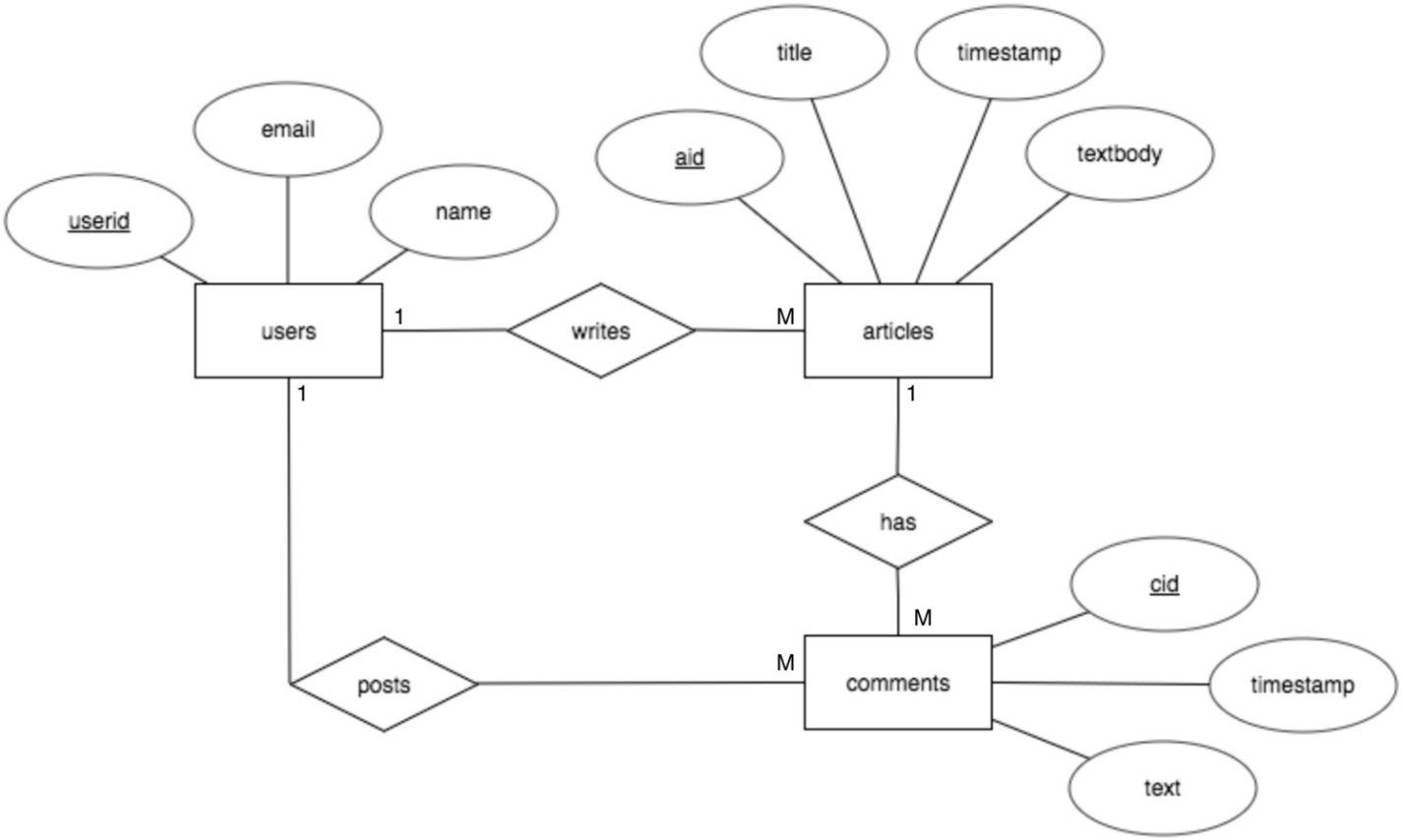
# Procedure of ERD

- Data modeling is iterative process.
- "complete" and "100% error free" model is not possible!
- Only "Optimized" model is possible….

**Steps to create an ERD**



# Sample ERD


title
timestamp
email
aid
textbody
userid
name
users
1
writes
M
articles
1
1
has
cid
M
M
posts
comments
timestamp
text

# Exercise

*Identify entities, relationships, and cardinalities in the below scenario of a university and create an ER diagram.*

Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course.

# Exercise

*Create an ERD for the following scenario*

The company is organized into DEPARTMENTs. Each department has a name, number and an employee who *manages* the department. We keep track of the start date of the department manager. Each department *controls* a number of PROJECTs. Each project has a name, number and is located at a single location. We store each EMPLOYEE's social security number, address, salary, sex, and birthdate. Each employee *works for* one department but may *work on* several projects. We keep track of the number of hours per week that an employee currently works on each project. We also keep track of the *direct supervisor* of each employee. Each employee may *have* a number of DEPENDENTs who are depending on the employees. For each dependent, we keep track of their name, sex, birthdate, and relationship to employee.

# End of the Session